δ _DELTA

MIGRATION

$$x'_{C,i} = \alpha_i X_i - \beta_i x_{C,i} x_{P,i} + \sum_{j=1,\, j\neq i}^{N_R} \gamma_{j,i} x_{C,j} - \sum_{j=1,\, j\neq i}^{N_R} \gamma_{i,j} x_{C,\mathrm{i}}$$

CRIMINAL

$$x'_{P,i} = \; - \delta_i x_{P,i} + \beta_i x_{C,i} x_{P,i} + \sum_{j=1,\, j\neq i}^{N_R} \theta_{i,j} x_{P,j} - \sum_{j=1,\, j\neq i}^{N_R} \theta_{j,i} x_{P,\mathrm{i}}$$

POLICE

Note: the indices j of each some are not related to each other, they just represent sums.

$$\gamma_{i,j} = \frac{x_{P,i} - x_{P,j}}{d(i,j)}$$

$$\gamma_{j,i} = \frac{x_{P,j} - x_{P,i}}{d(j,i)}$$

$$\theta_{i,j} = \frac{x_{C,i} - x_{C,j}}{d(i,j)}$$

$$\theta_{j,i} = \frac{x_{C,j} - x_{C,i}}{d(j,i)}$$

$d(i, j)$: distance between i-th and j-th region. $d(i, j) = d(j, i)$

$$x_{C,i} = \frac{Criminal\ Population\ the\ \ in\ i{-}th\ region}{Area\ of\ \ i{-}th\ region}$$

$$x_{P,i} = \frac{Police\ Population\ in\ the\ \ i{-}th\ region}{Area\ of\ \ i{-}th\ region}$$

<u>Arbitrary Numbers</u>:

N_P : Total number of police in all regions.

N_C : Total number of criminals in all regions.

<u>Variables:</u>

$x_{C,i}$: population density of criminals in the i-th region

$x_{P,i}$: population density of police in the i-th region

<u>Coefficients:</u>

α_i : HDI?

BETA β_i : criminal catching efficiency of police in the i-th region.

$\gamma_{j,i}$: frequency of criminals from region j leaving its region and going to region i.

$\gamma_{i,j}$: frequency of criminals from region i leaving its region and going to region j.

DELTA δ_i : the rate of natural police decrease: sum of police retirement, burnout, death, etc.

$\theta_{i,j}$: frequency of police from region i leaving its region and going to region j.

$\theta_{j,i}$: frequency of police from region j leaving its region and going to region i.

<u>Randomness:</u>

NOISE X_i: the random, impulse driven criminal growth rate. It is scaled by HDI α_i. This is because HDI and crime rates have strong linear correlation (0.7?)